

PIERCE: Perception-Integrated Execution in: Roboti Culinary Environments

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I. ABSTRACT

Bimanual manipulation of soft, contact-rich objects remains a significant challenge for low-cost robotic systems, particularly in culinary tasks such as food skewering, which require precise spatial alignment, coordinated dual-arm control, and reliable insertion under uncertainty. To address this problem, we propose PIERCE (Perception-Integrated Execution in Robotic Culinary Environments), an autonomous dual-arm robotic system mounted on a 360° rotating base. Without relying on expensive tactile sensors, the system uses multi-view RGB observations together with robot proprioception to perceive the workspace and guide manipulation. The core intelligence of PIERCE is a pretrained Vision-Language-Action (VLA) policy, implemented using Pi0.5 [3] and fine-tuned offline on expert demonstrations collected through a custom leader-follower teleoperation pipeline. By assigning complementary roles to the two arms, the system executes a closed-loop perception-action pipeline for grasping, alignment, and contact-rich insertion. We will evaluate the system using stage-wise and end-to-end metrics, including grasping success, alignment accuracy, insertion success, and full-task completion rate for assembling single-type fruit skewers. Ultimately, this project aims to demonstrate that a low-cost bimanual robot can adapt a modern VLA policy to perform precise and contact-rich culinary manipulation with reliable autonomous execution.

Keywords: Bimanual Manipulation, Vision-Language-Action, Teleoperation, Multi-View Perception, Contact-Rich Manipulation, Culinary Robotics.

II. INTRODUCTION

Preparing skewered food appears simple for humans, yet it poses a demanding manipulation problem for low-cost robotic systems. The task requires coordinated bimanual control, precise spatial alignment, and reliable execution of contact-rich insertion, all while handling soft and deformable objects such as fruit. Small errors in pose, timing, or force can easily lead to failure, for example by deforming the fruit, slipping during grasping, or missing the tip of the skewer during insertion. Existing automated culinary systems often rely on expensive tactile sensors, carefully engineered fixtures, or highly structured workspaces to manage these difficulties. However, for robotic assistance to become practical in domestic and small-scale culinary settings, there is a clear need for systems

that can perform such tasks using affordable hardware and perception-driven control.

Our proposed system, PIERCE (Perception-Integrated Execution in Robotic Culinary Environments), addresses this challenge through an autonomous dual-arm platform mounted on a 360° rotating base. The system uses multi-view RGB observations together with robot proprioception to perceive the workspace and guide manipulation, without depending on specialized tactile sensing. At the policy level, PIERCE leverages a pretrained Vision-Language-Action (VLA) model that is fine-tuned on expert demonstrations collected through a custom leader-follower teleoperation pipeline. This design enables complementary role assignment between the two arms: one arm stabilizes the skewer and workspace, while the other performs fruit grasping, alignment, and insertion. We focus on fruit skewering as a representative benchmark for low-cost culinary robotics because it concentrates several key challenges of real-world manipulation into a compact and measurable task.

The motivation for PIERCE lies in demonstrating that modern visuo-motor foundation policies can be adapted to affordable robotic platforms for precise, coordinated, and contact-rich execution. Rather than engineering a narrowly scripted solution, we seek to validate an end-to-end pipeline that transfers human demonstrations into reliable autonomous behavior. Success in this setting would lower the barrier to entry for culinary automation and provide a scalable blueprint for other household tasks that require multi-step coordination, perception under uncertainty, and fine physical interaction.

III. RELATED WORK

Recent work in culinary and bimanual robotics has shown that role decomposition is an effective strategy for handling deformable objects and coordination-heavy food preparation tasks. For example, the CARBS framework [1] uses a secondary arm as a physical constraint to stabilize ingredients during scooping, while prior work on robotic stir-frying [2] adopts a leader-follower architecture in which two arms perform complementary motions with strong temporal coordination. These systems suggest that assigning specialized functions to each limb can simplify control and improve robustness in manipulation settings that are difficult for a single arm to manage.

PIERCE System Architecture: SPA Loop & Imitation Learning Dataflow

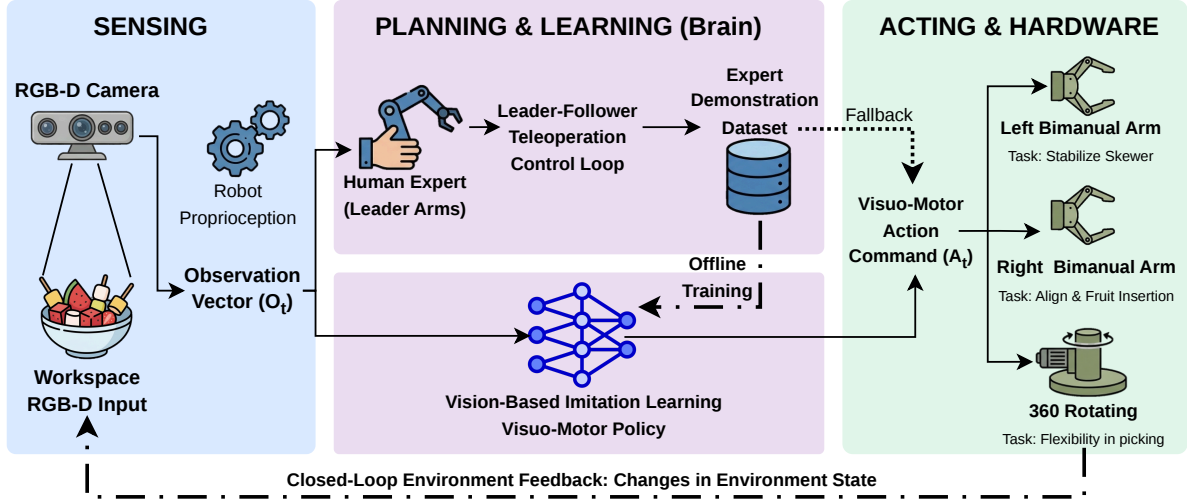


Fig. 1. PIERCE system architecture. The overall learning framework follows imitation learning from teleoperated demonstrations, while the policy backbone is implemented using the pretrained Pi0.5 Vision-Language-Action model.

Beyond task decomposition, recent progress in visuo-motor learning has made it increasingly practical to acquire complex manipulation behaviors from demonstrations rather than from hand-engineered task logic. In particular, teleoperation provides an efficient mechanism for collecting expert trajectories in bimanual settings, while modern Vision-Language-Action (VLA) policies offer a strong pretrained prior for mapping multimodal observations to continuous robot actions. This direction is especially relevant for low-cost robotic platforms, where the goal is to compensate for limited hardware sophistication through better use of visual observations, proprioception, and data-driven policy adaptation.

PIERCE builds on these lines of work while targeting a distinct and challenging manipulation setting: precise fruit skewering with low-cost hardware. Unlike scooping or stir-frying, skewering requires accurate axial alignment, stable dual-arm coordination, and reliable insertion into soft objects under geometric uncertainty. Our system therefore combines complementary bimanual roles with multi-view visual perception and offline fine-tuning of a pretrained VLA policy on teleoperated demonstrations. This design shifts the focus from scripted control toward an end-to-end perception-action pipeline capable of learning grasping, alignment, and insertion behavior directly from expert data. By addressing visual occlusion through a multi-camera setup and avoiding dependence on tactile sensing, PIERCE explores a practical path toward affordable culinary manipulation in real-world environments.

IV. PROPOSED METHOD

A. System Design and Hardware Architecture

The proposed PIERCE (Perception-Integrated Execution in Robotic Culinary Environments) system utilizes a dual-arm

robot mounted on a 360° rotating base to provide flexible access to objects across the workspace and improve approach angles for contact-rich manipulation. The hardware architecture is designed to avoid dependence on expensive tactile sensors, instead relying on multi-view visual perception and robot proprioception. The bimanual system assigns complementary roles to the two arms: the left arm stabilizes the skewer and holder, while the right arm performs fruit grasping, alignment, and insertion. For perception, the system integrates multiple RGB-D cameras with a 70° field of view to observe the workspace, detect fruit pieces, and localize the skewer holder from multiple viewpoints. This multi-camera setup helps reduce visual occlusion during close-proximity bimanual interaction and provides sufficient environmental coverage for autonomous execution.

B. Visuo-Motor Policy and Control Pipeline

Our overall learning framework follows imitation learning from teleoperated demonstrations, while the policy backbone is implemented using Pi0.5 [3], a pretrained Vision-Language-Action (VLA) model provided in the open-source LeRobot library [4]. At time t , the observation is defined as $\mathbf{o}_t = \{\mathbf{I}_t, \mathbf{q}_t, \mathbf{l}\}$, where $\mathbf{I}_t$ denotes multi-view visual observations, $\mathbf{q}_t$ denotes bimanual proprioceptive states, and $\mathbf{l}$ denotes the natural language task instruction. This formulation allows the policy to combine scene perception, robot state awareness, and task conditioning within a unified visuo-motor framework.

Instead of predicting a single low-level action at each step, Pi0.5 [3] uses Action Chunking to generate a short horizon sequence of continuous actions,

$$\mathbf{A}_t = [\mathbf{a}_t, \dots, \mathbf{a}_{t+k}],$$

which is well suited for smooth and coordinated dual-arm motion during grasping, alignment, and insertion. This is particularly important for skewering, where abrupt action changes can easily cause missed contact, fruit deformation, or unstable insertion. The policy is fine-tuned offline on expert demonstrations collected through our custom leader-follower teleoperation pipeline. In this sense, imitation learning serves as the training paradigm through which the pretrained VLA policy is adapted to our specific culinary manipulation task.

Following the Pi0.5 framework [3], policy adaptation is performed using its native training objective in LeRobot [4], which optimizes the model to generate action chunks that match expert behavior conditioned on observations. During autonomous deployment, the robot repeatedly updates these chunked actions in a closed-loop perception-action pipeline, allowing the system to respond to small workspace variations and maintain stable bimanual coordination throughout task execution.

C. Experimental Procedure

The experimental phase is structured into three stages: data collection, offline policy adaptation, and autonomous deployment. First, we will use the leader-follower teleoperation setup to collect expert demonstrations of fruit grasping, skewer stabilization, alignment, and insertion. These demonstrations will include both successful trajectories and, when feasible, recovery behavior under small variations in fruit placement and pose. Second, the collected dataset will be used to fine-tune the pretrained Pi0.5 policy offline. Third, the adapted policy will be deployed on the real robot to evaluate whether the system can autonomously assemble a simple single-type fruit skewer in a fixed tabletop workspace.

To ensure feasibility within the project timeline, the initial experiments will focus on a constrained setting with standardized fruit size, fixed camera placement, and a consistent skewer-holder configuration. We will first validate isolated subtasks such as grasp stability and insertion precision, and then evaluate complete end-to-end skewering performance across repeated trials. After establishing a reliable baseline, we will extend the evaluation to moderate workspace variation, including randomized fruit placement and small pose changes.

D. Foreseeable Challenges and Solutions

A primary challenge is the contact-rich nature of skewering soft food items, which requires precise alignment and coordinated insertion without tactile sensing. We address this by combining complementary bimanual role assignment with demonstration-driven policy learning, enabling the system to learn manipulation strategies directly from expert behavior rather than relying on manually engineered contact rules. To further improve reliability, we will begin with firmer fruit types and a constrained workspace before expanding to more deformable objects.

Another challenge is visual occlusion during close bimanual interaction. To mitigate this issue, the system uses multiple cameras and a rotating base to maintain broad workspace

visibility from different viewpoints. We also structure the manipulation sequence so that grasping and alignment occur in visually observable regions before insertion. In addition, low-cost hardware may introduce backlash, calibration drift, or unstable motions. We therefore plan to use conservative motion speeds, repeated calibration, and safety checks in the control loop so that execution can be halted if predicted motions exceed safe operating conditions.

A final challenge is the distribution gap between teleoperated demonstrations and autonomous execution. To reduce this gap, we will collect demonstrations under small variations in object placement and workspace state, and, when possible, include corrective trajectories that expose the policy to minor recovery behavior. This helps improve the robustness of the imitation learning pipeline when transferring from expert demonstrations to autonomous deployment.

E. Reach Goals

Once baseline reliability for single-type fruit skewering is established, future work will focus on increasing the system’s dexterity and interactivity. One direction is to generalize the learned manipulation policy to multiple food types with different textures and geometries, such as marshmallows, grapes, oranges, bananas, and watermelon. Another direction is to develop a user interface that allows users to specify skewer composition, enabling the robot to assemble multi-item skewers in a user-defined order.

V. PROPOSED EVALUATION

Our evaluation framework measures whether PIERCE can reliably assemble a simple single-type fruit skewer under a constrained tabletop setup using both quantitative metrics and qualitative assessment. To ensure that system performance is analyzed in a structured way, we evaluate the robot progressively, from isolated subtasks to full end-to-end autonomous execution.

A. Quantitative Metrics

• Stage 1: Single-Item Grasping Reliability

- *Metric:* Successful Pickup Rate.
- *Procedure:* The robot attempts to grasp a single grape placed at randomized positions within the workspace.
- *Success Criteria:* A trial is counted as successful if the grape is lifted from the surface and held securely without slipping, dropping, or being visibly crushed.

• Stage 2: Single-Item Insertion Precision

- *Metric:* Skewering Success Rate.
- *Procedure:* Starting from a successful grasp, the robot must align the fruit with the skewer held by the stabilizing arm and complete insertion.
- *Success Criteria:* A trial is counted as successful if the fruit is fully pierced and remains stably positioned on the skewer without falling or deforming excessively.

• Stage 3: End-to-End Task Success

- *Metric*: Full Task Completion Rate.
- *Procedure*: The robot must autonomously locate, grasp, align, and skewer a sequence of grapes from start to finish.
- *Success Criteria*: A trial is counted as successful if the robot assembles a complete 3-grape skewer without human intervention or safety interruption.

- **Alignment Quality**

- *Metric*: Alignment Accuracy.
- *Procedure*: During insertion trials, we will record whether the fruit is aligned with the skewer axis before contact.
- *Success Criteria*: Higher performance corresponds to fewer failed insertion attempts caused by lateral misalignment or missed contact.

- **Execution Efficiency**

- *Metric*: Average Task Completion Time.
- *Procedure*: We measure the time required for the autonomous policy to complete one successful skewering cycle and compare it against the average expert teleoperation time.
- *Success Criteria*: Lower cycle time, provided that task success and safety are maintained.

B. Qualitative Assessments

- **System Robustness**: We will observe how well the policy handles small workspace variations, including moderate changes in fruit placement, orientation, and visual appearance.
- **Bimanual Coordination Quality**: We will qualitatively assess whether the stabilizing arm and insertion arm exhibit smooth and complementary behavior during grasping, alignment, and insertion.
- **Failure Modes**: We will document common failure cases, such as grasp instability, visual occlusion, missed alignment, or fruit deformation, to guide future system improvements.

C. Expected Outcome

We expect to deliver a reliable end-to-end visuo-motor system that demonstrates the feasibility of adapting a pretrained Vision-Language-Action (VLA) policy to low-cost bimanual culinary manipulation. More specifically, we expect PIERCE to achieve consistent grasping, alignment, and insertion performance for assembling simple single-type fruit skewers under a constrained workspace setup.

VI. TIMELINE

Our project timeline prioritizes our base goal automating our robot to use bimanual manipulation to stack 3 grapes on a single wooden skewer with 95% accuracy. Going into the week of March 22, we have printed and fully assembled the robot arms. We have also printed the robot base but haven’t tested its stability with the arms attached.

Week	Goals
3/27	Finalize design, assemble the robot, verify basic arm motion and camera setup.
4/3	Set up teleoperation and perception, and collect initial demonstrations.
4/10	Train the first imitation learning policy and test single-item skewering.
4/17	Integrate dual-arm control, perception, and autonomous skewering in a fixed workspace.
4/19	Improve reliability & begin stretch goals if progress allows
4/24	Improve reliability and begin stretch goals.
4/27	Final demonstration and report submission.

TABLE I
PROJECT TIMELINE

VII. BUDGET

Our project budget will be primarily put in cameras, fruit pieces for the robot to handle, and the suction mount securing the robot to tables and counter tops. We were able to source the servos for the robot arms and spinning base from the Robot Garage hosted by the University of Michigan Robotics Department and our personal stashes. The robot arms will be printed on the 3D printers in the Ford Robotics Building and on our personal 3D printers, so we estimate no cost for 3D printing. The estimated breakdown of our budget for this project is described in Table II below.

Item	Quantity	Unit Cost (\$)	Total Cost (\$)
RGB Cameras	4	20	80
Skewers & Assorted Fruit	1	30	30
Suction Cup Mount	1	40	40
Total			150

TABLE II
PROJECT BUDGET

VIII. REFERENCES

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